Effect of climate change interventions on pear quality using Ecotron units
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Experimental Design and Sample Size Calculation

PMHD Project - Group 10

Apu Chandra Bhowmick
Basma Lusta
Olurotimi Koleowo
Omar Ben Gadha

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Introduction

- Climate change poses increasing challenges to agricultural productivity and food quality worldwide.
- Controlled-environment studies allow researchers to simulate future climate conditions and understand their impact on crops.
- This project uses Ecotron technology to simulate different climate scenarios and study their effects on pear production in Belgium.

Research Question and Objectives

- **Research Question:** Which climate scenario improves pear quality the most?
- **Overall Objective:** Study different climate change interventions on fruit production in Belgium.
- **Study Objective:** Identify the most favorable climate intervention for the best pear quality.

Study design

- **Control group:** Climate scenario 1.
- **Intervention groups:** Climate scenarios 2, 3, 4.
- **Blinding:** Single-blind — quality assessors were unaware of the climate scenario per Ecotron.
- **Randomization:** Stratified block randomization was used: climate scenarios were randomly assigned to Ecotron units, with equal pear species (3 trees per species) within each unit.
- **Duration/End of study:** Continue harvesting until the last harvestable pear is harvested.
- **Experimental units:** The harvested pears.

Hypothesis Test

- **Null Hypothesis (H_0):**
  There is no significant difference in mean pear quality between climate intervention scenarios. The effect of climate change interventions on pear quality are the same/equal.

$$H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4$$

- **Alternative Hypothesis (H_1):**
  At least one climate intervention mean scenario significantly improves pear quality compared to the control scenario.

$$H_1 : \mu_i > \mu_1 \text{ for at least one i} \qquad (i = 2, 3, 4)$$

Statistical Model — Linear Mixed Effects Model (LMM)

$$Y_{ki} = \beta_0 + \beta_1 * (Scenario)_{2k} + \beta_2 * (Scenario)_{3k} + \beta_3 * (Scenario)_{4k} + \beta_4 * (Variety)_k + b_k + \varepsilon_{ki}$$

Where:

- Y_{ki} = Quality score of pear i from tree k
- β_0 = Control groups (climate scenario 1 and pear variety: Conference)
- β_1, β_2, β_3 = Fixed effects of different climate intervention scenarios compared to the control
- β_4 = Fixed effect of pear variety
- b_k = Random effect for trees $N(0, \sigma^2_k)$
- ε_{ki} = Residual error i.i.d $N(0, \sigma^2_{ki})$

Pilot Data Summary

- Purpose: to estimate the mean and standard deviation (SD) of pear quality to inform sample size calculations
- Total pears: 591
- Source: Collected from Ecotron units under climate scenario 1 (Control group)
- Varieties: Conference & Doyenne du Comice
- Quality scale: 0 to 100

Parameters for Sample Size Calculation

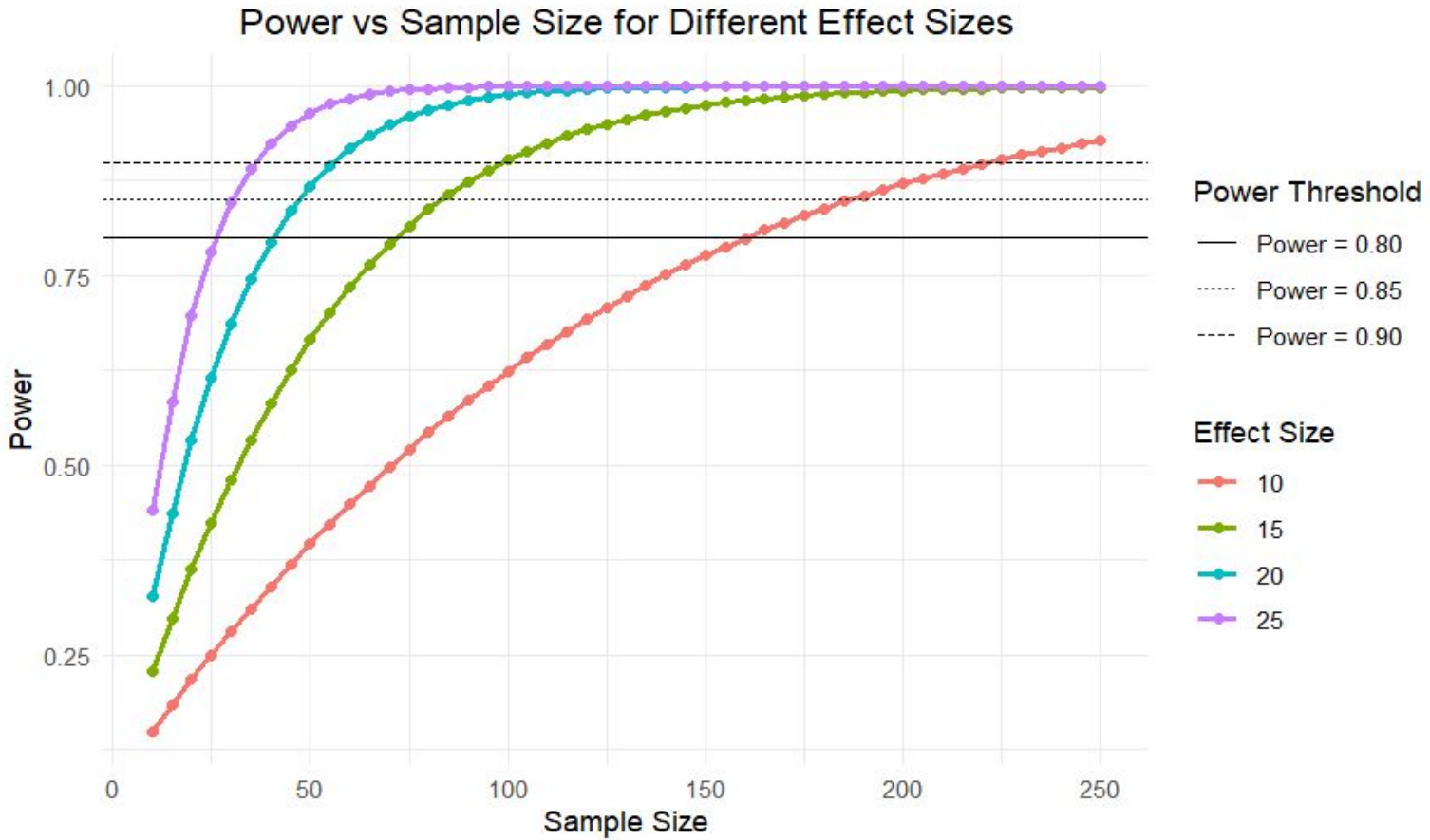
- Significance level (α): 0.05
- Mean value : 51.08
- Standard Deviation: 35.9
- Effect size: from 10 to 25
- Power level: from 80% to 90%

Sample Size Estimation Methods

1. Power Test-Based Sample Size Calculation Method

- Used R’s power.t.test() function
- One-sided two-sample t-test
- Inputs:
  - Effect size (10 -25)
  - P-value : 0.05
  - Standard deviation from pilot data
  - Target power (80% - 90%)
- Output
  - required sample size per group

Visualization of Sample Size Calculation



Power Test-Based Sample Size Table

Table 1: the sample size of different level of power, effect size (10-25), and significance level of (0.05)

Table 1: Sample Size Calculation

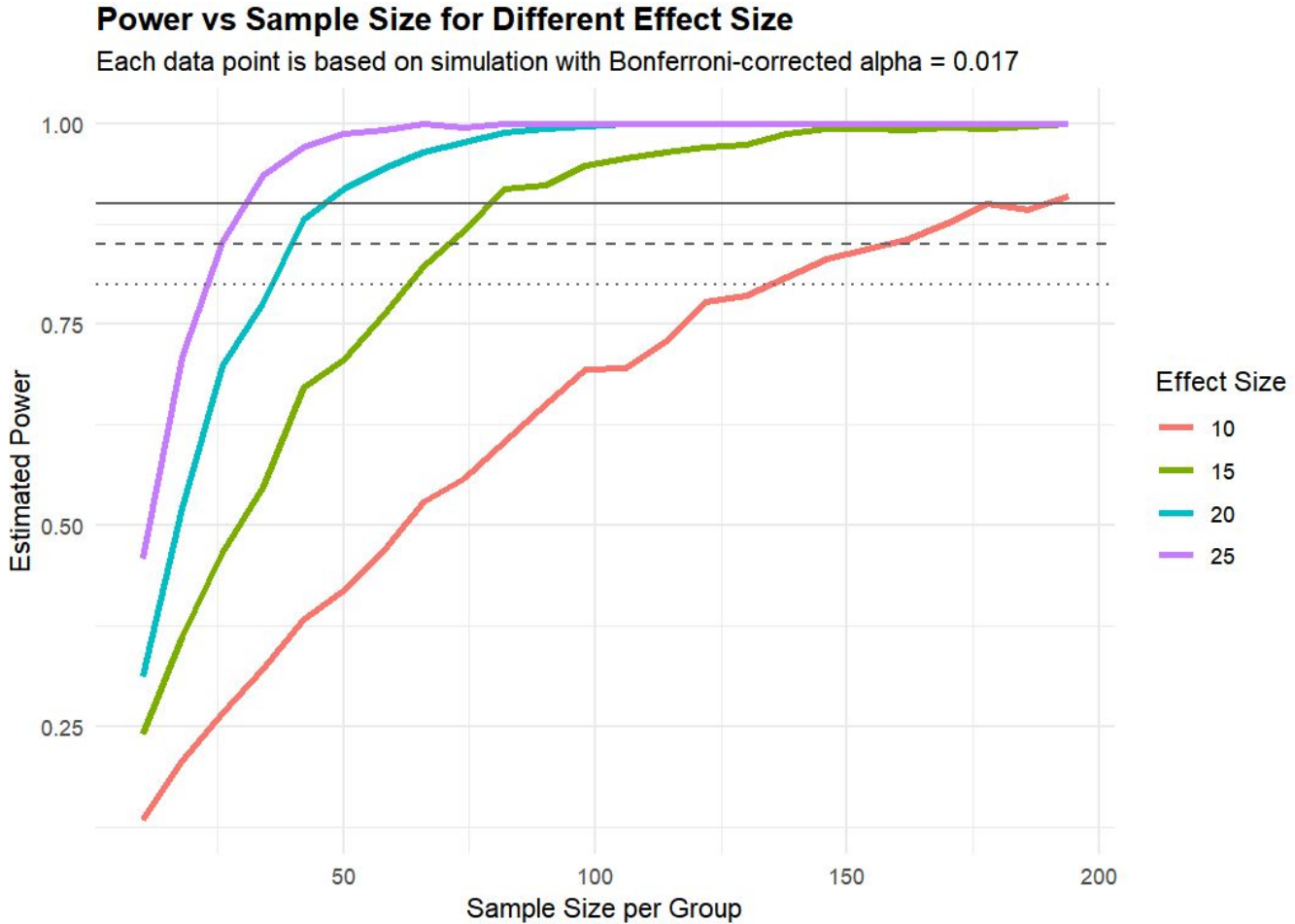
Power(δ)	Effect Size	Significance Level(α)	Sample Size(N)
0.80	10	0.05	160.39
0.85	10	0.05	186.39
0.90	10	0.05	221.91
0.80	15	0.05	71.67
0.85	15	0.05	83.23
0.90	15	0.05	99.00
0.80	20	0.05	40.62
0.85	20	0.05	47.12
0.90	20	0.05	55.99
0.80	25	0.05	26.26
0.85	25	0.05	30.41
0.80	25	0.05	36.09

Sample Size Estimation Methods

2- Simulation-Based Sample Size Calculation Method

- Simulated data for Control and 3 Climate Scenarios
- Quality scores assumed to follow a normal distribution.
  - Mean and standard deviation taken from pilot data
- For each simulation:
  - Performed three one-sided t-tests (each scenario vs. control)
  - Applied Bonferroni correction for multiple comparisons (0.05 / 3)
  - Counted as a success if at least one scenario significantly outperformed control
- Estimated power = Proportion of successful simulations

Visualization of Sample Size Calculation



Simulation-Based Sample Size Table

- Table 2 : Estimated Sample Sizes by Effect Size and Power (Bonferroni α = 0.017)

Table 2: Simulation-Based Sample Size Calculation

Power(δ)	Effect Size	Significance Level(α)	Sample Size(N)
0.809	10	0.017	138
0.832	10	0.017	146
0.844	10	0.017	154
0.856	10	0.017	162
0.876	10	0.017	170
0.900	10	0.017	178
0.893	10	0.017	186
0.822	15	0.017	66
0.867	15	0.017	74
0.881	20	0.017	42
0.853	25	0.017	26

Ecotron Capacity & Allocation

- 12 Ecotron units available in Belgium.
- 12 optional Ecotron units available in France if needed.
- Each Ecotron unit can contain 6 trees at maximum.
- Each tree produces 30–40 pears per yield.
- Therefore, each unit yields:
  - Minimum: 6 × 30 = 180 pears.
  - Maximum: 6 × 40 = 240 pears.

Summary

- **Goal:** Identify the climate intervention scenario that results in the highest pear quality under controlled Ecotron conditions.
- **Sample Size Estimate**: 160–220 pears per group depends on methods (power t-test or simulation methods).
- **Next Steps:** Finalize Ecotron unit allocation and confirm design with sponsors and professors.

Thank you for your attention